

Mathematical Methods for Finance

Module 2 – Measure, Measurable Functions and First Measures

Lesson 6: From sets to measurable objects

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3-hour lesson structure: intuition, theory, examples, exercises, and exam-style practice

Learning outcomes for Lesson 6

By the end of this lesson, students should be able to:

- ① explain why sigma-algebras are needed before defining integration;
- ② construct small sigma-algebras and recognize Borel sets;
- ③ verify the basic properties of measures in finite examples;
- ④ check measurability of simple functions;
- ⑤ translate ordinary set operations into the language of measurable spaces.

Exam relevance

This lesson provides the language needed to read the next modules: measurable sets, measurable functions and measures are the base layer of integration and probability.

How today is organised

14:30–15:20

Concepts and intuition

From arbitrary sets to admissible sets. Sigma-algebras, Borel sets.

15:35–16:25

Theory and examples

Measures.

16:40–17:30

Exercises

Small sigma-algebras, finite measures and measurable functions.

Intuition: Learning rule

For each topic: identify the object, write the definition, test it on a small example, then solve one exam-style exercise.

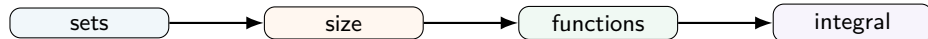
Module 2

Block 1: why measurable sets? Sigma-algebras and Borel sets

Module 2 in one sentence

Intuition: The central idea

Module 2 builds a rigorous language for assigning size to sets and integrating functions.



Course strategy

We keep the theory precise, but every definition will be tested with a finite or interval-based example.

Counting is not enough

Example: Finite set

If $X = \{a, b, c, d\}$, the size of $A = \{a, c\}$ can be counted: $\#A = 2$.

Example: Interval

If $A = [0, 1]$, it contains infinitely many points. Counting does not distinguish $[0, 1]$ from $[0, 2]$.

Intuition: Need for measure

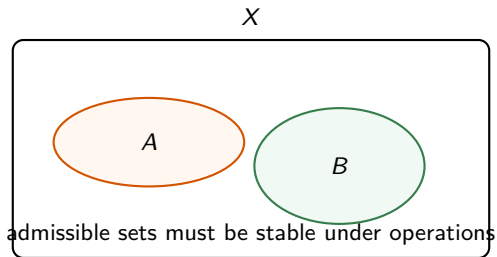
A measure generalizes size: it can count points, measure lengths, and later assign probabilities.

What is allowed to be measured?

Intuition: A key warning

Before assigning size, we must decide which subsets are admissible.

- In finite examples, usually every subset is admissible.
- In continuous spaces, allowing all subsets creates deep technical problems.
- The solution is to work with a family of admissible sets: a sigma-algebra.



Sigma-algebra: definition

Definition: Sigma-algebra

Let X be a set. A family $\mathcal{F} \subseteq \mathcal{P}(X)$ is a sigma-algebra if:

- ① $X \in \mathcal{F}$;
- ② if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- ③ if $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

The pair $(X, \mathcal{F})$ is called a measurable space.

Intuition: Meaning

If a set is admissible, then its complement and countable unions of admissible sets are admissible.

Why complements and countable unions?

Complement

If A is observable, then “not A ” should also be observable.

Union

If A_n are observable, then “at least one A_n happens” should be observable.

Intersection

Intersections come for free by De Morgan:

$$\cap_n A_n = (\cup_n A_n^c)^c.$$

Exercise: Quick check **

Starting from the axioms, prove that $\emptyset \in \mathcal{F}$.

Solution: why the empty set belongs to a sigma-algebra

Proof idea: One line

Since $X \in \mathcal{F}$ and sigma-algebras are stable under complements,

$$\emptyset = X^c \in \mathcal{F}.$$

Exam lesson

Many proofs in measure theory are short once the correct axiom is identified.

The smallest and largest sigma-algebras

Example: Two extremes

On any set X :

$$\{\emptyset, X\}$$

is the smallest sigma-algebra, while

$$\mathcal{P}(X)$$

is the largest sigma-algebra.

Intuition: Meaning

The smallest sigma-algebra observes only impossible/certain events. The power set observes every subset.

Finite sigma-algebra example

Worked example: A four-point space

Let $X = \{1, 2, 3, 4\}$ and

$$\mathcal{F} = \{\emptyset, X, \{1, 2\}, \{3, 4\}\}.$$

Then $\mathcal{F}$ is a sigma-algebra.

- X and $\emptyset$ are present.
- Complements stay in $\mathcal{F}$: $\{1, 2\}^c = \{3, 4\}$.
- Unions of elements of $\mathcal{F}$ are still in $\mathcal{F}$.

Exercise: is this a sigma-algebra?

Exercise: Finite test **

Let $X = \{1, 2, 3\}$ and

$$\mathcal{F} = \{\emptyset, X, \{1\}, \{2, 3\}\}.$$

Is $\mathcal{F}$ a sigma-algebra?

Method: Steps

- 1 Check X and $\emptyset$.
- 2 Check complements.
- 3 Check unions.

Solution: finite test

Proof idea: Detailed steps

We have $X \in \mathcal{F}$ and $\emptyset \in \mathcal{F}$. Also

$$\{1\}^c = \{2, 3\}, \quad \{2, 3\}^c = \{1\}.$$

All unions among the four sets remain one of the four sets. Therefore $\mathcal{F}$ is a sigma-algebra.

Counterexample: not a sigma-algebra

Counterexample: Missing complement

Let $X = \{1, 2, 3\}$ and

$$\mathcal{G} = \{\emptyset, X, \{1\}\}.$$

Then $\mathcal{G}$ is not a sigma-algebra because

$$\{1\} \in \mathcal{G}, \quad \{1\}^c = \{2, 3\} \notin \mathcal{G}.$$

Lesson

To disprove a sigma-algebra property, one missing axiom is enough.

Generated sigma-algebra

Definition: Sigma-algebra generated by a family

Given a family $\mathcal{A} \subseteq \mathcal{P}(X)$, the sigma-algebra generated by $\mathcal{A}$ is

$\sigma(\mathcal{A})$ = the smallest sigma-algebra containing $\mathcal{A}$.

Example: Finite example

If $X = \{1, 2, 3, 4\}$ and $\mathcal{A} = \{\{1, 2\}\}$, then

$$\sigma(\mathcal{A}) = \{\emptyset, X, \{1, 2\}, \{3, 4\}\}.$$

Borel sigma-algebra

Definition: Borel sets

The Borel sigma-algebra on $\mathbb{R}$, denoted $\mathcal{B}(\mathbb{R})$, is the sigma-algebra generated by the open intervals of $\mathbb{R}$.

Intuition: Meaning

Borel sets are the sets that can be constructed from intervals using countable unions, countable intersections and complements.

Example: Examples

Open intervals, closed intervals, half-lines, singletons and countable sets are Borel sets.

Why intervals generate Borel sets



Intuition: Mental picture

Intervals are the elementary building blocks of the real line. Borel sets are obtained by repeatedly combining these blocks.

Exercise: Quick check **

Show that $\{x\}$ is a Borel set.

Solution: singletons are Borel

Proof idea: Use closed intervals

For any $x \in \mathbb{R}$,

$$\{x\} = \bigcap_{n=1}^{\infty} \left(x - \frac{1}{n}, x + \frac{1}{n}\right).$$

Each interval is open, hence Borel. Countable intersections of Borel sets are Borel. Therefore $\{x\}$ is Borel.

Measurable spaces

Definition: Measurable space

A measurable space is a pair $(X, \mathcal{F})$ where X is a set and $\mathcal{F}$ is a sigma-algebra on X .

Example: Finite

$X = \{1, 2, 3\}$ with $\mathcal{F} = \mathcal{P}(X)$.

Example: Real line

$(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Key point

A measurable space says which sets are admissible. It does not yet assign sizes to them.

Module 2

Block 2: measures

Measure: definition

Definition: Measure

Let $(X, \mathcal{F})$ be a measurable space. A measure is a function

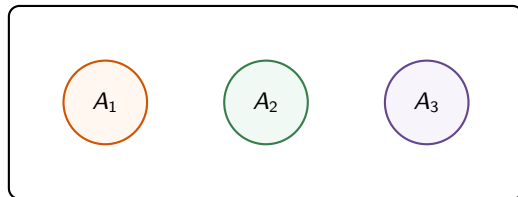
$$\mu : \mathcal{F} \rightarrow [0, +\infty]$$

such that:

- ① $\mu(\emptyset) = 0$;
- ② if $(A_n)_{n \geq 1}$ are pairwise disjoint sets in $\mathcal{F}$, then

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) = \sum_{n=1}^{\infty} \mu(A_n).$$

The meaning of countable additivity



Intuition: Disjoint pieces

If sets do not overlap, the size of their union is the sum of their sizes.

$$\mu(A_1 \cup A_2 \cup A_3) = \mu(A_1) + \mu(A_2) + \mu(A_3).$$

Three fundamental examples of measures

Example: Counting measure

$$\mu(A) = \#A.$$

It counts elements.

Example: Lebesgue measure

$$\lambda([a, b]) = b - a.$$

It measures length.

Example: Probability

$$P(X) = 1.$$

It measures likelihood.

Bridge

Probability theory is measure theory with total mass one.

Finite measure example

Worked example: Weights on points

Let $X = \{1, 2, 3\}$ and $\mathcal{F} = \mathcal{P}(X)$. Define weights

$$w_1 = 2, \quad w_2 = 5, \quad w_3 = 1.$$

For $A \subseteq X$, define

$$\mu(A) = \sum_{i \in A} w_i.$$

Then μ is a measure.

Exercise: Compute *

Find $\mu(\{1, 3\})$ and $\mu(X)$.

Solution: finite measure example

Proof idea: Computation

$$\begin{aligned}\mu(\{1, 3\}) &= w_1 + w_3 = 2 + 1 = 3. \\ \mu(X) &= w_1 + w_2 + w_3 = 2 + 5 + 1 = 8.\end{aligned}$$

Lesson

On a finite space, a measure is just a consistent way of adding weights.

Probability as a special measure

Definition: Probability measure

A measure P on $(\Omega, \mathcal{F})$ is a probability measure if

$$P(\Omega) = 1.$$

The triple $(\Omega, \mathcal{F}, P)$ is a probability space.

Example: Finite probability

If $\Omega = \{\omega_1, \omega_2, \omega_3\}$ and probabilities are $(0.2, 0.3, 0.5)$, then

$$P(\{\omega_1, \omega_3\}) = 0.2 + 0.5 = 0.7.$$

Basic properties of measures

Theorem: Consequences of countable additivity

For a measure μ :

- ① If $A \subseteq B$, then $\mu(A) \leq \mu(B)$.
- ② If $A \subseteq B$ and $\mu(A) < \infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$.
- ③ $\mu(A \cup B) \leq \mu(A) + \mu(B)$.

Exercise: Short proof **

Prove monotonicity: if $A \subseteq B$, then $\mu(A) \leq \mu(B)$.

Solution: monotonicity

Proof idea: Split the larger set

If $A \subseteq B$, then

$$B = A \cup (B \setminus A),$$

and the union is disjoint. Hence

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A).$$

Therefore $\mu(A) \leq \mu(B)$.

Null sets

Definition: Null set

A set $N \in \mathcal{F}$ is null if

$$\mu(N) = 0.$$

Example: Lebesgue measure

For Lebesgue measure on $\mathbb{R}$,

$$\lambda(\{x\}) = 0$$

for every point x .

Intuition: Important idea

A set can contain points and still have measure zero.

Almost everywhere

Definition: Almost everywhere

A property holds μ -almost everywhere if it fails only on a null set.

Example: Functions equal almost everywhere

Let

$$f(x) = 0 \quad \text{for all } x, \quad g(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

Then $f = g$ almost everywhere with respect to Lebesgue measure.

Module 2

Block 3: measurable functions

Measurable functions: definition

Definition: Measurable function

Let $(X, \mathcal{F})$ be a measurable space. A function $f : X \rightarrow \mathbb{R}$ is measurable if, for every Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$f^{-1}(B) = \{x \in X : f(x) \in B\} \in \mathcal{F}.$$

Intuition: Meaning

A measurable function is one whose observable output events correspond to admissible input sets.

A more usable measurability criterion

Theorem: Half-line criterion

A function $f : X \rightarrow \mathbb{R}$ is measurable if and only if

$$\{x \in X : f(x) > a\} \in \mathcal{F}$$

for every $a \in \mathbb{R}$.

Why useful?

Instead of checking every Borel set, we check all upper-level sets.

Indicator functions

Definition: Indicator

For a set $A \subseteq X$,

$$\mathbf{1}_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Theorem: Measurability

$\mathbf{1}_A$ is measurable if and only if $A \in \mathcal{F}$.

Intuition: Why it matters

Indicators are the elementary bricks used to build simple functions and then the Lebesgue integral.

Simple functions

Definition: Simple function

A simple function is a measurable function taking finitely many values. It can be written as

$$s(x) = \sum_{k=1}^m a_k \mathbf{1}_{A_k}(x),$$

where $A_k \in \mathcal{F}$.

Example: Finite values

If $A, B \in \mathcal{F}$ are disjoint, then

$$s = 3\mathbf{1}_A - 2\mathbf{1}_B$$

is simple.

Measurability exercise

Exercise: Indicator and simple functions **

Let $(X, \mathcal{F})$ be a measurable space and $A, B \in \mathcal{F}$. Show that

$$f = 2\mathbf{1}_A + 5\mathbf{1}_B$$

is measurable.

Method: Shortcut

Use the fact that indicators of measurable sets are measurable, and finite linear combinations of measurable functions are measurable.

Solution: indicator and simple functions

Proof idea: Step-by-step

Since $A, B \in \mathcal{F}$, $\mathbf{1}_A$ and $\mathbf{1}_B$ are measurable. Multiplying by constants preserves measurability. Adding measurable functions preserves measurability. Therefore

$$f = 2\mathbf{1}_A + 5\mathbf{1}_B$$

is measurable.

Continuous functions are Borel measurable

Theorem: Continuity implies measurability

If $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then f is Borel measurable.

Proof idea: Idea

For every open set $O \subseteq \mathbb{R}$, continuity gives that $f^{-1}(O)$ is open. Open sets are Borel. Hence $f^{-1}(O)$ is Borel.

Example: Examples

Polynomials, exponentials, logarithms on their domains, trigonometric functions and absolute value are Borel measurable.

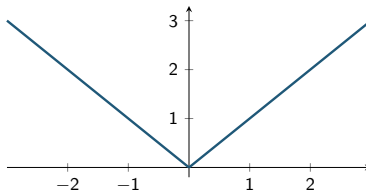
Absolute value is a measurable function

Example: A function used constantly

The function

$$f(x) = |x|$$

is continuous, hence Borel measurable.



Intuition: Bridge to Module 1 and 2

Absolute value measures distance on $\mathbb{R}$ and builds norms, inequalities and integrable magnitudes.

Counterexample mentality

Counterexample: A function cannot be measurable without specifying a sigma-algebra

The sentence “ f is measurable” is incomplete unless we know the sigma-algebra on the domain and codomain.

Example: Finite warning

Let $X = \{1, 2, 3\}$ and $\mathcal{F} = \{\emptyset, X\}$. A nonconstant function $f : X \rightarrow \mathbb{R}$ is typically not measurable because some inverse images are nontrivial subsets not in $\mathcal{F}$.

Exam workshop: finite measurable function

Exercise: Finite space ***

Let $X = \{1, 2, 3, 4\}$ and

$$\mathcal{F} = \{\emptyset, X, \{1, 2\}, \{3, 4\}\}.$$

Define $f(1) = f(2) = 0$ and $f(3) = f(4) = 1$. Is f measurable?

Solution: finite measurable function

Proof idea: Look at level sets

The function takes only values 0 and 1. The relevant inverse images are

$$f^{-1}(\{0\}) = \{1, 2\} \in \mathcal{F}, \quad f^{-1}(\{1\}) = \{3, 4\} \in \mathcal{F}.$$

All inverse images of Borel sets are obtained by combining these sets. Hence f is measurable.

End of Lesson 6: what to remember

Checklist

After this lesson you should be able to answer:

- ① What is a sigma-algebra?
- ② What is a Borel set?
- ③ What is a measure?
- ④ What does “almost everywhere” mean?
- ⑤ How do we check measurability in simple examples?

Exercise: Suggested practice

From the exercise pack: sigma-algebras, measures and measurable functions. Start from the lowest difficulty and write the definition before solving.